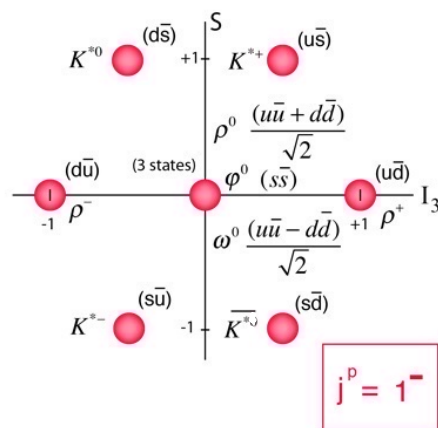
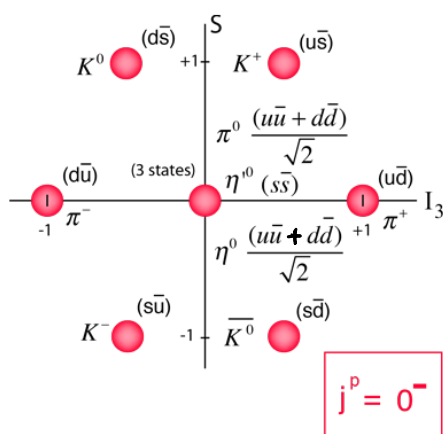


Meson multiplets

$$K^0 \rightarrow K^+ \pi^-$$



SU(3) Flavor, static quark model

$$\text{meson} = q_i \bar{q}_j$$

$$q_i = u, d, s$$

$$\bar{q}_j = \bar{u}, \bar{d}, \bar{s}$$

$$M = 3 \otimes \bar{3} = 1 \oplus 8$$

1: Flavor Singlet

8: Flavor octet

$$q_i \bar{q}_j$$

$$\uparrow\uparrow$$

$$\downarrow\downarrow$$

$$S=1$$

$$S_z = \pm 1$$

$$J = L + S = 1$$

$$\uparrow\downarrow$$

$$\downarrow\uparrow$$

$$S=0$$

$$S_z = 0$$

$$\Rightarrow J = L + S = 0$$

mesons in the ground state: $L=0$

3 states/particles: $I_3=0$, Strangeness $=0 \Rightarrow Q=0$

$$\psi_{8,1} = (u\bar{u} - d\bar{d})/\sqrt{2}$$

$$\psi_{8,0} = (u\bar{u} + d\bar{d} - 2s\bar{s})/\sqrt{6}$$

$$\psi_1 = (u\bar{u} + d\bar{d} + s\bar{s})/\sqrt{3}$$

observe particles for {pseudoscalar} octets
vector

$\Rightarrow$ study decays $\Rightarrow I_3$ for the particles

$\Rightarrow$ which physical particles assign
to $\psi_{8,1}, \psi_{8,0}, \psi_1$

$$f = \psi_8 \cos \theta - \psi_1 \sin \theta$$

$$f' = \psi_8 \sin \theta + \psi_1 \cos \theta$$

θ mixing angle
 between $I=1$
 and $I=0$ states
 in the octet

π^0, η, η'
 are observed
 physical
 states

$I=1$ pion triplet

$$\left. \begin{aligned}
 \pi^0(140) &\approx \psi_{8,1}^{ps} = (u\bar{u} - d\bar{d})/\sqrt{2} \\
 \eta(550) &= \psi_{8,0}^{ps} \cos \theta_{ps} - \psi_1^{ps} \sin \theta_{ps} \\
 \eta'(960) &= \psi_{8,0}^{ps} \sin \theta_{ps} + \psi_1^{ps} \cos \theta_{ps}
 \end{aligned} \right\} \begin{aligned}
 J^P &= 0^-, \\
 \theta_{ps} &\approx -25^\circ
 \end{aligned}$$

$\pi^0: S=0$
 $\rho^0: S=1$
 quark content:
 $u\bar{d}, \bar{u}d$

$$\left. \begin{aligned}
 \rho^0(770) &\approx \psi_{8,1}^v = (u\bar{u} - d\bar{d})/\sqrt{2} \\
 \phi(1020) &= \psi_{8,0}^v \cos \theta_v - \psi_1^v \sin \theta_v \approx s\bar{s} \\
 \omega(780) &= \psi_{8,0}^v \sin \theta_v + \psi_1^v \cos \theta_v \approx \\
 &\approx (u\bar{u} + d\bar{d})/\sqrt{2}
 \end{aligned} \right\} \begin{aligned}
 J^P &= 1^-, \\
 \theta_{\text{vect}} &\approx 36^\circ
 \end{aligned}$$

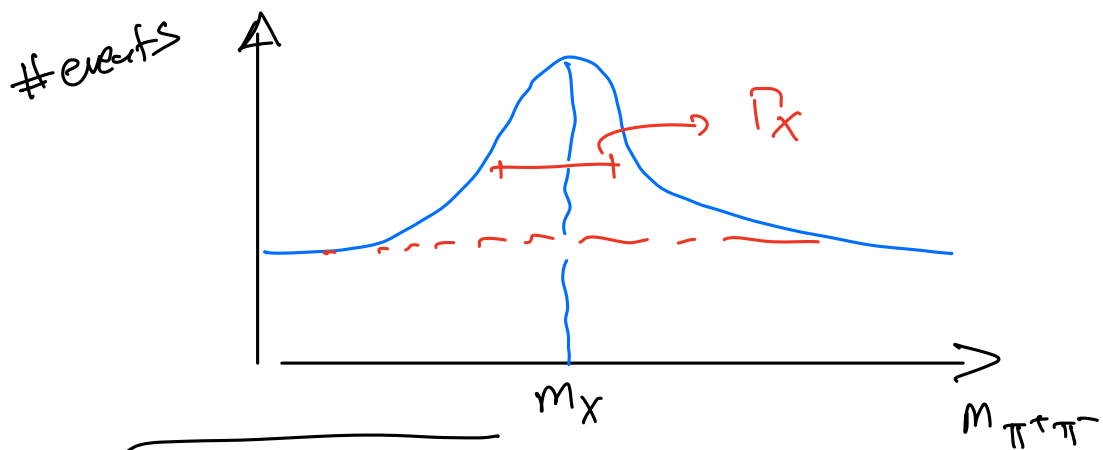
$L=0$ }
 $L=1$ }
 $L=2$ }

$n^{2s+1}\ell_J$	J^{PC}	$I=1$ $u\bar{d}, \bar{u}d,$ $\frac{1}{\sqrt{2}}(d\bar{d} - u\bar{u})$	$I=\frac{1}{2}$ $u\bar{s}, d\bar{s};$ $\bar{d}s, \bar{u}s$	$I=0$ f	$I=0$ f'
1^1S_0	0^{-+}	π	K	η	$\eta'(958)$
1^3S_1	1^{--}	$\rho(770)$	$K^*(892)$	$\phi(1020)$	$\omega(782)$
1^1P_1	1^{+-}	$b_1(1235)$	K_{1B}^a	$h_1(1415)$	$h_1(1170)$
1^3P_0	0^{++}	$a_0(1450)$	$K_0^*(1430)$	$f_0(1710)$	$f_0(1370)$
1^3P_1	1^{++}	$a_1(1260)$	K_{1A}^a	$f_1(1420)$	$f_1(1285)$
1^3P_2	2^{++}	$a_2(1320)$	$K_2^*(1430)$	$f_2'(1525)$	$f_2(1270)$
1^1D_2	2^{-+}	$\pi_2(1670)$	$K_2(1770)^a$	$\eta_2(1870)$	$\eta_2(1645)$
1^3D_1	1^{--}	$\rho(1700)$	$K^*(1680)^b$	$\phi(2170)^d$	$\omega(1650)$
1^3D_2	2^{--}		$K_2(1820)^a$		
1^3D_3	3^{--}	$\rho_3(1690)$	$K_3^*(1780)$	$\phi_3(1850)$	$\omega_3(1670)$
1^3F_4	4^{++}	$a_4(1970)$	$K_4^*(2045)$	$f_4(2300)$	$f_4(2050)$
1^3G_5	5^{--}	$\rho_5(2350)$	$K_5^*(2380)$		
2^1S_0	0^{-+}	$\pi(1300)$	$K(1460)$	$\eta(1475)^c$	$\eta(1295)$
2^3S_1	1^{--}	$\rho(1450)$	$K^*(1410)^b$	$\phi(1680)$	$\omega(1420)$
2^3P_1	1^{++}	$a_1(1640)$			
2^3P_2	2^{++}	$a_2(1700)$	$K_2^*(1980)$	$f_2(1950)$	$f_2(1640)$

$p+p \rightarrow p+p + (\text{X})$, contains many mesons

at increasing energy of proton $\Rightarrow$ heavier X
 in final state

$$p+p \rightarrow p+p + \pi^+ \pi^- \pi^+ \pi^- \pi^+ \pi^- \pi^0 \pi^-$$



$$m_{\pi^+\pi^-} = \sqrt{(\underline{p}_{\pi^+} + \underline{p}_{\pi^-})^2}$$

$$\underline{p}_{\pi^+} = (E_{\pi^+}, \vec{p}_{\pi^+})$$

$$E_{\pi^+} = \sqrt{m_{\pi}^2 + |\vec{p}_{\pi^+}|^2}$$

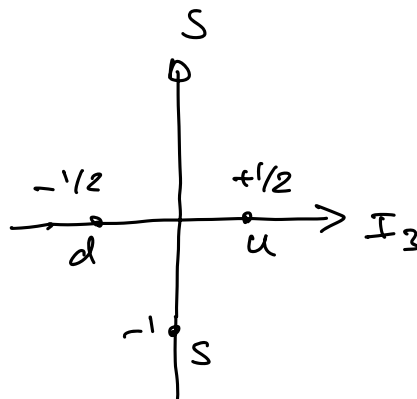
measure $|\vec{p}_{\pi^+}|$ from spectrometer

$$P = qBR$$

$$q = ze$$

$$P [\text{GeV}] = 0.3 \times z \times B(\tau) \times R(\text{m}).$$

Baryons
 $q_1 q_2 q_3$



uuc, uud, udd, ddd
 uus, uss, ssu

}

27 possible combinations.

$$\text{Baryon: } 3_F \otimes 3_F \otimes 3_F$$

3: fundam. rep of $SU(3)_F$

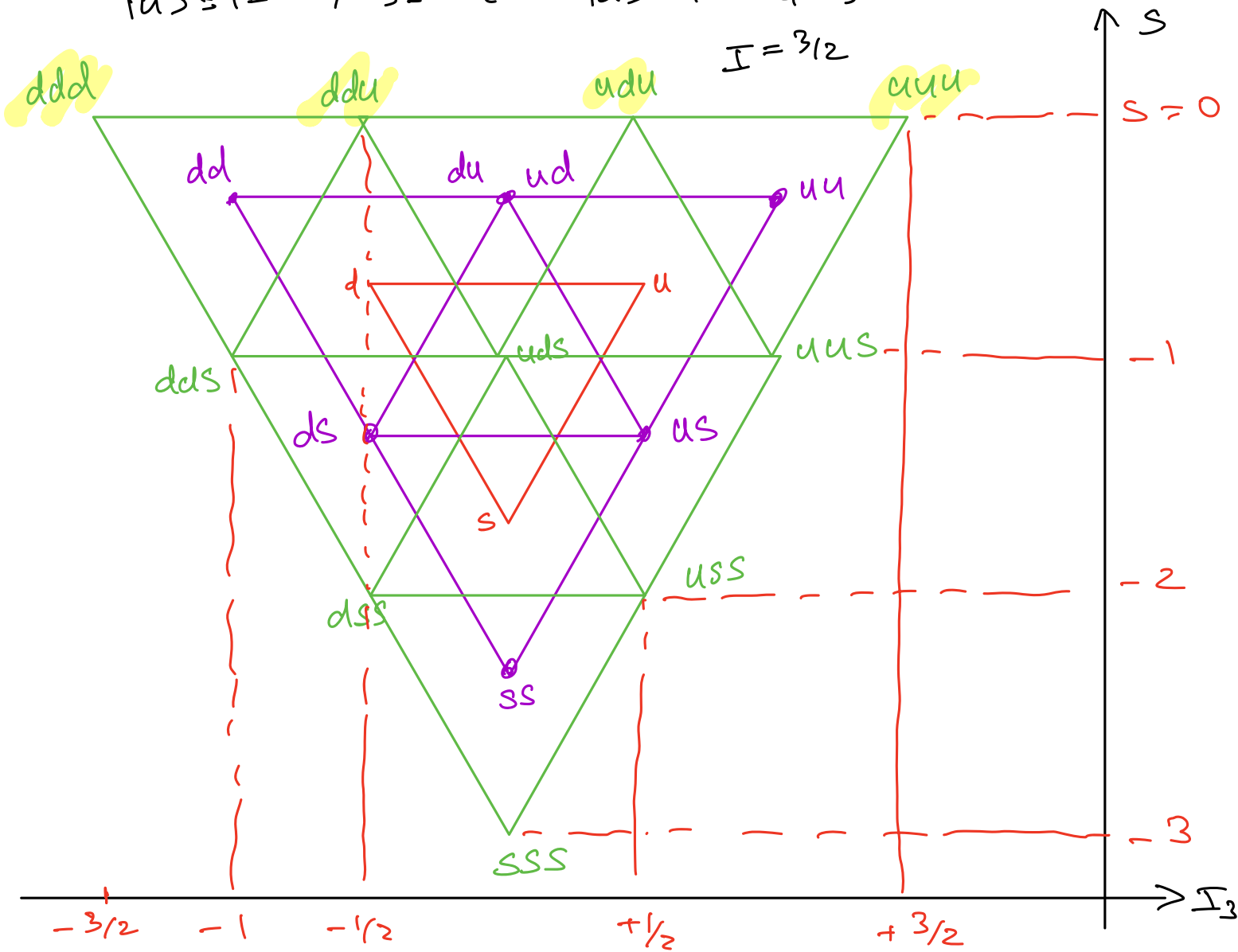
$$u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad d = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad s = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$B = 3_F \otimes 3_F \otimes 3_F = 10_F \oplus 8_{F,12} \oplus 8_{F,123} \oplus 10_F$$

F_{12} : $q_1 q_2 q_3$
 sym. under exchange of q_1, q_2

F_{123} : $q_1 q_2 q_3$
 sym. under exchange of $2 \leftrightarrow 3$

$$|u\rangle = |I=1/2, I_3 = +1/2\rangle \quad |d\rangle = |I=1/2, I_3 = -1/2\rangle \quad S=0$$



Spin Configurations for baryons:

1st Combo

$$a_1 \quad a_2 \quad a_3$$

$$S=1/2 \quad \uparrow\downarrow \quad \uparrow\downarrow \quad \uparrow\downarrow$$

$$\frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} =$$

$$\uparrow\uparrow\uparrow \quad \downarrow\downarrow\downarrow \quad S=3/2$$

$$J^P = 3/2^+$$

ground state $L=0 \Rightarrow J = L+S = 3/2$

$$P = P_1 P_2 P_3 (-1)^L = (+1)^3 (-1)^0 = +1$$

2nd Combo

$$\uparrow\uparrow\downarrow \quad \uparrow\downarrow\uparrow \quad \downarrow\uparrow\uparrow$$

$$S_z = +1/2, -1/2 \Rightarrow S=1 \Rightarrow J=1/2$$

$L=0$ ground state

$\Rightarrow$

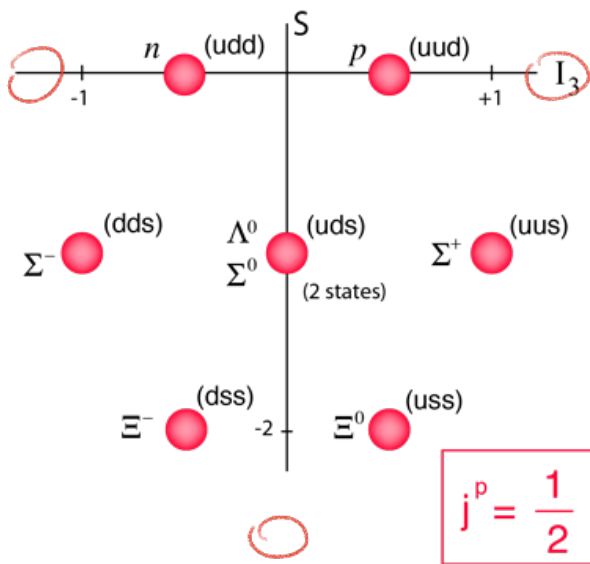
$$J^P = 1/2^+$$

$$P = +1$$

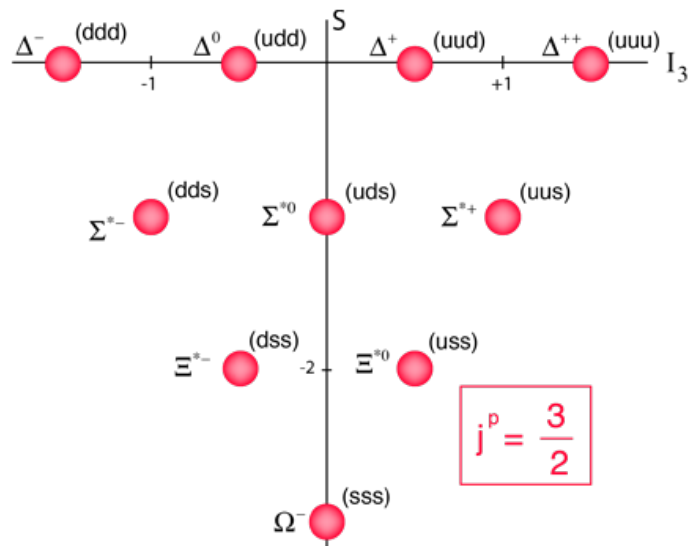
$\Rightarrow$ no C eigenstates

because $q \neq \bar{q}$

Expected baryons : 27 ($3 \times 3 \times 3$)



octet



decuplet

Baryons : fermions.

$$\Delta^{++} : S = 3/2 \quad uuu$$

$$u \uparrow \quad u \uparrow \quad u \uparrow$$

$$\Delta^- : d \uparrow \quad d \uparrow \quad d \uparrow$$

$$\Omega^- : s \uparrow \quad s \uparrow \quad s \uparrow$$

$$\psi_{\text{baryon}} = \psi_{\text{space}} \psi_{\text{flavor}} \psi_{\text{spin}}$$

$$(-1)^L \quad S \quad S$$

Symm.

$\Rightarrow$ symm. wave function for baryons
which are fermions.

Introduce color as a new charge/degree of freedom.

$$\psi_{\Delta^{++}} = \psi_{\text{space}} \psi_{\text{flav}} \psi_{\text{spin}} \psi_{\text{color}}$$

S S S A

A: antisymm. under exchange
of any pair of quarks.

$$\Delta^{++} \quad \uparrow u_R \quad \uparrow u_R \quad \uparrow u_R$$

$\Rightarrow$ 1 color cannot work

$$\Delta^{++} \quad \uparrow u_R \quad \uparrow u_B \quad \uparrow u_R$$

A₁₂

A₂₃

S₁₃

} Cannot be Antisymm.
for all exchanges.

$\Rightarrow$ at least 3 colors.

$$\Delta^{++} \quad \uparrow u_R \quad \uparrow u_G \quad \uparrow u_B$$

$$\psi_{\text{color}} = [RGB + BRG + GBR + \\ - GRB - RBG - BGR] / \sqrt{6}$$

completely antisymm. under $1 \leftrightarrow 2, 2 \leftrightarrow 3, 1 \leftrightarrow 3$

Baryon wave functions

$$\psi_B = \psi_{\text{space}} \psi_{\text{spin}} \psi_{\text{flavor}} \psi_{\text{color}}$$

S S A

$$L=0 \Rightarrow \psi_{\text{space}} \text{ Symm.}$$

$$S=1/2 \quad \psi_{\text{spin}}^{S=1/2} \quad A$$

$$S=3/2 \quad \psi_{\text{spin}}^{3/2} \quad S$$

$$3_F \otimes 3_F \otimes 3_F = \mathbb{1}_{F, \text{Symm.}} \oplus 8_{F, M_{12}} \oplus 8_{F, M_{23}} \oplus \mathbb{1}_{F, A}$$

Symm. under any exchange Symm. 1↔2 Symm. 2↔3

$$3_C \otimes 3_C \otimes 3_C = \mathbb{1}_{C, S} + 8_{C, 12} + 8_{C, 23} + \mathbb{1}_{C, A}$$

Hypothesis: physical particles are all color singlets $\mathbb{1}_{C, A}$

Decuplet

$$\psi_{10} = \psi_{\text{space}} \psi_{\text{spin}}^{3/2} \psi_{\text{flavor}} \psi_{\text{color}}$$

S S S A

Flavor Singlet

$$\psi_{F, 1} = \psi_{\text{space}} \psi_{\text{spin}} \psi_{\text{flavor}} \psi_{\text{color}}$$

S S A A

↗ $\mathbb{1}_{F, A}$

Symm. total wave function

$$\text{Flavor Symm.} \quad \left[uds + sud + dsu + \dots \right] / \sqrt{6}$$

$$\text{Flavor anti-symm.} \quad \left[uds + sud + dsu + \dots - usd - sdu - dus \right] / \sqrt{6}$$

Flavor Octets

$$8_{F, M_{12}}$$

$$8_{F, M_{23}}$$

$$S = 1/2 \quad \frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} = \left(\frac{3}{2}\right)_S + \left(\frac{1}{2}\right)_{M_{12}} + \left(\frac{1}{2}\right)_{M_{23}}$$

$$\psi_{\text{octet}} = \underbrace{\psi_{\text{space}}}_S \underbrace{\psi_{\text{spin}} \psi_{\text{Flavor}} \psi_{\text{color}}}_{\text{S}} \underbrace{\psi_{\text{color}}}_A$$

$$S = 3/2 \Rightarrow \psi_{\text{spin}} = \text{Symm. always.}$$

$$\psi_{\text{Flavor}}: \text{Symm. under } 1 \leftrightarrow 2, \text{ or } 2 \leftrightarrow 3$$

$$\Rightarrow \text{no octet with } S = 3/2 \text{ can be always Symm.}$$

$$3_F \otimes 3_F \otimes 3_F = \Pi_{F,S} + \underbrace{8_{F, M_{12}} + 8_{F, M_{23}}}_{\text{not always Symm.}} + \Pi_{F,A}$$

$$\Rightarrow S = 3/2 \text{ Flavor-octet } \alpha \text{ cannot be always Symm.}$$

$$\alpha(\text{ways}) \equiv 1 \leftrightarrow 2, 2 \leftrightarrow 3, 1 \leftrightarrow 3$$

$$\Rightarrow 8 \quad S = 3/2 \text{ octet baryons not physical states.}$$

$\Rightarrow$ possibility left $S=1/2$ Flavor octets.

$$S=1/2 \quad \psi_{F_8} = \psi_{\text{space}} \psi_{\text{spin}} \psi_{\text{Flavor}} \psi_{\text{color}}$$

$S \qquad \qquad \qquad A$

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} =$$

$$\left(\frac{3}{2}\right)_S + \left(\frac{1}{2}\right)_{S,12} + \left(\frac{1}{2}\right)_{S,23}$$

$$\psi_{\text{spin},12}^{1/2}$$

$$\psi_{\text{Flavor},12}$$

$$\psi_{\text{spin},23}^{1/2}$$

$$\psi_{\text{Flavor},23}$$

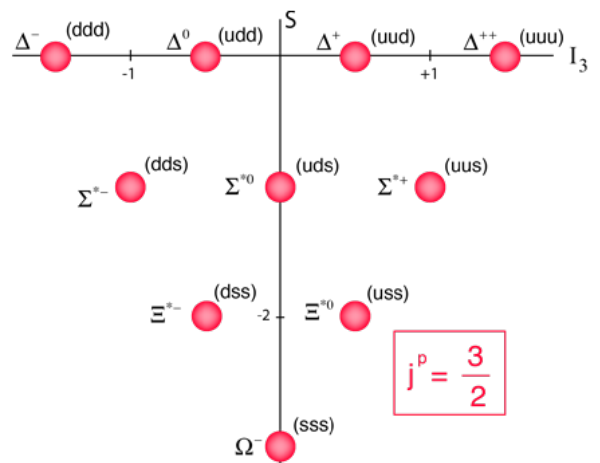
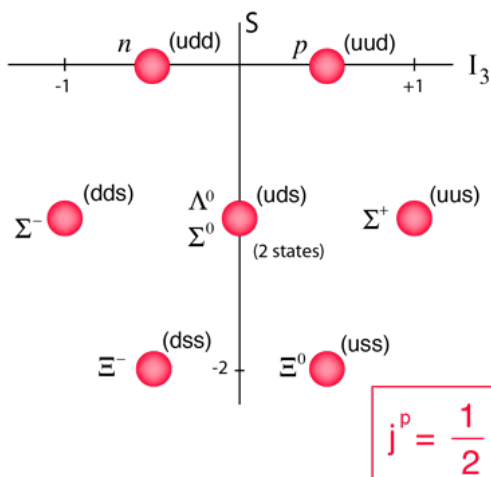
$\Rightarrow$ Flavor octet corresponds to $F_{12} \times S_{12} + F_{23} \times S_{23}$

From 27 baryons:

1 Flavor singlet not physical.

8 Flavor with $S=3/2$ not physical.

$\Rightarrow$ 18 physical states.



Meson wave function

$$M = q_i \bar{q}_j \quad 3_F \times \bar{3}_F = 1_F + 8_F$$

$$\psi_{\text{meson}} = \psi_{\text{space}} \underbrace{\psi_{\text{spin}} \psi_{\text{flavor}}}_{\substack{S \\ L=0 \\ 1 \leftrightarrow 2 \\ ?}} \psi_{\text{color}} \quad \text{A hypothesis.}$$

$q_i, \bar{q}_j$ not identical particles.

$$\pi = u \bar{d} \quad \Delta^{++} = \uparrow_u \uparrow_u \uparrow_u$$

$1 \leftrightarrow 2 \quad \bar{d}u \quad \uparrow_u \uparrow_u \uparrow_u$

How to have color singlet for mesons:

3 colors: R, G, B $R = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad G = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$SU(3)_{\text{color}}$

quarks have color R, G, B .

anti-quarks have anti-color $\bar{R}, \bar{B}, \bar{G}$

$$R + \bar{R} = 0 = B + \bar{B} = G + \bar{G}$$

$$R + B + G = 0$$

$$3_C \otimes \bar{3}_C = 1_{CS} + 8_{CA}$$

Color singlet
for mesons.

8 color objects
mediators of
Quantum Chromodynamics
gluons.